

CSCE 748 COMPUTATIONAL PHOTOGRAPHY · MODULE 08 · LECTURE NOTES

Gradient-Domain Processing

Editing derivatives instead of values.

MODULE OUTCOMES

Explain why humans perceive gradients rather than absolute values.

Derive the Poisson equation from a least-squares gradient fit.

Implement seamless cloning.

Apply gradient-domain methods beyond compositing.

Solve the resulting linear system efficiently.

1 Why gradients

THE VISUAL SYSTEM MEASURES DIFFERENCES, NOT LEVELS

Retinal ganglion cells have centre-surround receptive fields — excited by light in the centre and inhibited by light in the surround, or the reverse. **The output is a local contrast, not an intensity.**

Absolute brightness is largely discarded at the first stage of processing.

This is why simultaneous contrast illusions work. Two patches of identical grey look markedly different against light and dark surrounds, because what is reported is the difference and the difference genuinely differs.

And why you can read a page in direct sunlight and in a dim room, despite a factor of a thousand in absolute luminance. The contrast between ink and paper is roughly constant, and that is what is being measured.

So if you preserve the gradients, the result looks right — even if every absolute value in the image has changed. **That is the entire premise of this module,** and it is why a technique that discards absolute values can nonetheless produce convincing images.

A SEAM IS A GRADIENT DISCONTINUITY

Pasting a region from one image into another creates a visible boundary because the values jump at the edge — a large gradient where the underlying scene has none. The eye, which is measuring gradients, detects it immediately.

So reframe the operation. Instead of copying the source's *values*, copy its *gradients*, and require the result to match the destination exactly along the boundary.

Then no discontinuity exists at the boundary by construction, because the boundary values are the destination's own.

The interior absolute values change — the pasted region takes on the destination's overall brightness and colour cast, since that is what the boundary conditions propagate inward. **And that is exactly what you want:** a face pasted from a sunlit photograph into a shaded scene *should* become shaded. The method does the colour matching for free, as a consequence of its structure rather than as an additional step.

2 The Poisson equation

$$\min_f \iint_{\Omega} \|\nabla f - \mathbf{v}\|^2 \quad \text{subject to} \quad f = f^* \text{ on } \partial\Omega$$

Find the function f over the region Ω whose gradient most closely matches a guide field $\mathbf{v}$, in a least-squares sense, subject to f matching the destination image f^* on the region's boundary. **The boundary conditions are what make the solution unique** — without them, any constant could be added to f .

$$\Rightarrow \nabla^2 f = \nabla \cdot \mathbf{v}$$

The Euler–Lagrange condition for that variational problem is the **Poisson equation**: the Laplacian of the unknown equals the divergence of the guide field. **The derivation matters** — it establishes that this is a principled least-squares fit rather than a procedure that happens to work.

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For each INTERIOR pixel p:
  |N(p)|*f(p) - SUM_{q in interior} f(q)
    = SUM_{all q} v(p,q) + SUM_{q on boundary} f*(q)
                                ^ known destination values
  where v(p,q) = g(p) - g(q)   from the SOURCE image

A x = b with A sparse, symmetric, positive definite (a Laplacian),
x the unknown interior values, b the divergence plus boundary terms.
One system per colour channel.
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THIS IS THE SAME SYSTEM AS THE FLUID PRESSURE SOLVE

CSCE 649 Module 11 solved $\nabla^2 \mathbf{p} = \nabla \cdot \mathbf{u}^*$ to project a velocity field onto the divergence-free subspace.

This module solves $\nabla^2 f = \nabla \cdot \mathbf{v}$ to integrate a gradient field into an image.

Same matrix, same solvers, same scaling behaviour. The matrix is the graph Laplacian of the pixel grid: five non-zeros per row, symmetric, positive definite. Conjugate gradient with a preconditioner works; multigrid is asymptotically optimal.

Poisson problems recur throughout computational science — electrostatics, heat flow, incompressible flow, image integration — and recognising one saves reinventing its solution. The decades of work on multigrid solvers were not done for image editing, and image editing benefits from all of it.

3 Applications

Guide field v	Resulting tool
The source image's gradients	Seamless cloning. The source's content appears, taking on the destination's lighting.
At each pixel, whichever of the source and destination gradients is larger	Mixed seamless cloning. See below.
Source gradients, scaled down	A softened insertion — the source's content appears faintly, as though printed on the destination.
Zero everywhere	Object removal. With no gradient to match, the solution is the smoothest function satisfying the boundary — a membrane interpolating inward from the surrounding content. The simplest form of inpainting.
Destination gradients with the colour channels altered	Local recolouring that respects the existing structure.
The image's own gradients with large ones attenuated	Gradient-domain tone mapping (Module 04).

One solver, many tools. The guide field is the entire interface, and the elegance of that is worth noticing: a single piece of numerical machinery supports compositing, inpainting, recolouring, and tone mapping, and the difference between them is a few lines constructing v .

MIXED GRADIENTS PRESERVE WHAT IS UNDERNEATH

Plain seamless cloning replaces the destination entirely within the region — whatever texture was there is overwritten, because the guide field comes only from the source.

Mixed cloning takes, at each pixel and in each direction, whichever gradient has the larger magnitude — the source's or the destination's.

So strong destination texture survives. Where the destination has a prominent edge and the source is smooth, the destination's gradient wins and the edge is retained; where the source has content and the destination is flat, the source wins.

This is how you paste handwriting onto a brick wall and have the brick texture show through the strokes, or paste an object onto a textured surface and have the texture read through it. **It is a one-line change to the guide field construction** and the result looks disproportionately impressive.

4 Limits and solving

- **Very different colours.** Forcing the boundary to match the destination propagates inward, so a dark object pasted into a bright scene is brightened throughout — sometimes to the point of washing out. **The method matches lighting, which is usually desirable and occasionally destructive.**

- **Strong structure crossing the boundary that does not continue.** If an edge in the source meets the boundary and has no counterpart in the destination, the solution smears to reconcile the two, producing a visible smudge.
- **Large regions.** The boundary's influence decays with distance, so the interior of a very large pasted region can drift away from both images.
- **Mismatched texture, perspective, or scale. No amount of boundary matching fixes a composite that is wrong for other reasons** — the result is seamless and obviously fake, which is a distinctive and instructive failure. **The method removes seams; it does not create plausibility.**
- **And it is not fast.** Each edit requires a global solve over the region, which is why the next callout matters for anything interactive.

SOLVING IT FAST ENOUGH TO BE INTERACTIVE

Direct sparse Cholesky factorisation works well for modest regions, and the factorisation can be reused if the region's shape is unchanged — so dragging a pasted object around is cheap after the first solve.

Preconditioned conjugate gradient is the standard iterative choice, since the matrix is symmetric positive definite. An incomplete Cholesky preconditioner substantially reduces the iteration count.

Multigrid is asymptotically optimal — $O(n)$ rather than the $O(n^{1.5})$ of preconditioned CG on this problem — and it is what makes megapixel-scale gradient-domain editing interactive.

Or approximate. Mean-value coordinates give a closed-form interpolation of the boundary difference that is visually very close to the true Poisson solution at a small fraction of the cost, requires no solver at all, and is what several interactive tools actually ship. **An approximation that is visually indistinguishable is not a compromise** when the criterion is perceptual.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Pérez, Gangnet & Blake — Poisson Image Editing (free)	The paper. Seamless cloning, mixed gradients, and the variants of §3, in eight pages. One of the most immediately useful graphics papers ever written.
CMU 15-463 — gradient-domain processing lecture and assignment (free)	The derivation and the implementation. The assignment is Project 2's third option.
Bhat et al. — GradientShop: A Gradient-Domain Optimization Framework (free)	The general framework of §3 — many image operations as one gradient-domain optimisation.
Farbman et al. — Coordinates for Instant Image Cloning (free)	Mean-value coordinates — the fast approximation in §4.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Demonstrate simultaneous contrast with two identical grey patches on different backgrounds, and measure that the pixel values are equal.
- 2 Paste a region naively and observe the seam. Measure the gradient magnitude at the boundary.
- 3 Derive the discrete Poisson system for a small region by hand and confirm the matrix structure.
- 4 Implement seamless cloning with a conjugate gradient solver. Paste a face from a sunlit photograph into a shaded one and observe the colour adaptation.
- 5 Implement mixed gradient cloning and paste text onto a textured surface. Compare against plain cloning.
- 6 Set the guide field to zero and perform object removal. Compare against a modern inpainting tool.
- 7 Implement gradient-domain tone mapping (Module 04) using the same solver and confirm the machinery is unchanged.
- 8 Construct a failure case: paste something at the wrong scale and perspective, and observe that the result is seamless and obviously wrong.
- 9 Compare solve times for direct Cholesky, preconditioned CG, and a multigrid solver at several region sizes. Plot the scaling.
- 10 Implement mean-value coordinate cloning and compare against the true Poisson solution, both visually and numerically.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Why does the visual system respond to gradients, and what two everyday observations follow?
- 2 Why is a seam a gradient discontinuity, and what does fixing the boundary achieve?
- 3 Derive the Poisson equation from the least-squares gradient fit.
- 4 What is the structure of the discrete system, and what other problem shares it?
- 5 Name six guide fields and the tool each produces.
- 6 What do mixed gradients do, and give an example.
- 7 Give five situations where gradient-domain compositing fails.
- 8 Name four ways to solve the system, and say which is asymptotically optimal.